

Causal Analysis of Student Outcomes

Module AAA_2013J — Open University Learning Analytics Dataset (OULAD)

Preliminary Analysis — June 2026

1. Overview

This is a preliminary causal analysis of student academic outcomes in the AAA_2013J module from the Open University Learning Analytics Dataset. The goal is to understand whether VLE engagement and assessment performance causally affect whether students end up Withdrawn, Fail, Pass, or Distinction — and if so, which student subgroups are affected most.

The outcome variable (`final_result`) is encoded as an ordinal scale: Withdrawn=0, Fail=1, Pass=2, Distinction=3. Treatments are total VLE clicks and five assessment scores (1752–1756) in chronological order within the module. Seven confounders are controlled for: age band, gender, IMD band, highest education, disability, region, and prior attempts.

The pipeline runs in three stages: DoWhy for causal identification and ATE estimation, EconML CausalForestDML for heterogeneous treatment effects (CATE), and SHAP on the CATE model to identify which student characteristics drive treatment effect heterogeneity.

2. Causal Structure (DAG)

The DAG below shows the assumed causal structure. All confounders (pink nodes) have directed edges into both the treatment and the outcome, meaning they simultaneously influence a student's engagement/assessment performance and their final result. Conditioning on these confounders blocks all non-causal paths, satisfying the backdoor criterion for identification.

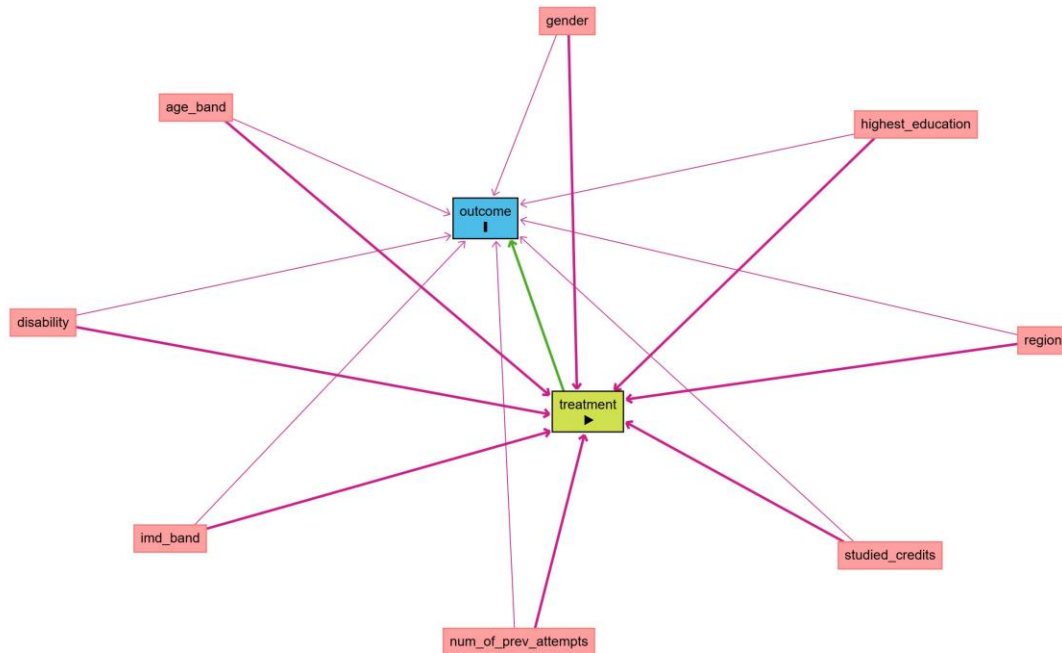


Figure 1: Causal DAG — confounders (pink) point to both treatment and outcome. Green arrow = causal path of interest.

Note: the DAG uses a generic "treatment" node for readability. In practice, six separate causal models are estimated — one for total_vle_clicks and one for each of the five assessment scores.

3. Average Treatment Effects (DoWhy)

Six separate causal models were estimated using linear regression with backdoor adjustment. Each model controls for all 36 encoded confounder variables. Three refutation tests were run per treatment: random common cause, subset data, and placebo. All six treatments passed all three tests.

3.1 VLE Clicks

ATE = 0.000146 per click. This is a tiny per-click effect, but across the mean click volume of 1,693 clicks, it translates to a cumulative shift of about 0.25 ordinal units. That's roughly a quarter of the gap between Withdrawn and Fail — modest but causally real and robustly estimated. The low heterogeneity (CATE std = 0.000043) suggests VLE engagement benefits all students fairly uniformly.

3.2 Assessment Scores

All five assessment scores show positive and significant ATEs. Across the full score range (0–100), each assessment causally shifts expected outcome by approximately 1.7 to 2.3 ordinal units — practically spanning the difference between Withdrawn and Pass, or between Fail and Distinction.

The progression is important: ATE increases from 1752 (earliest, ATE=0.017) through to 1756 (final, ATE=0.022). This means early assessment performance has a somewhat weaker causal link to final outcome compared to later assessments. Students who struggle early still have room to recover; by the final assessments, performance is a near-decisive driver of outcome category.

4. Heterogeneous Treatment Effects (EconML CausalForestDML)

CausalForestDML was used to estimate individual-level CATEs — how much the treatment effect varies across students with different characteristics. A separate model was fitted for each treatment.

4.1 CATE Distributions

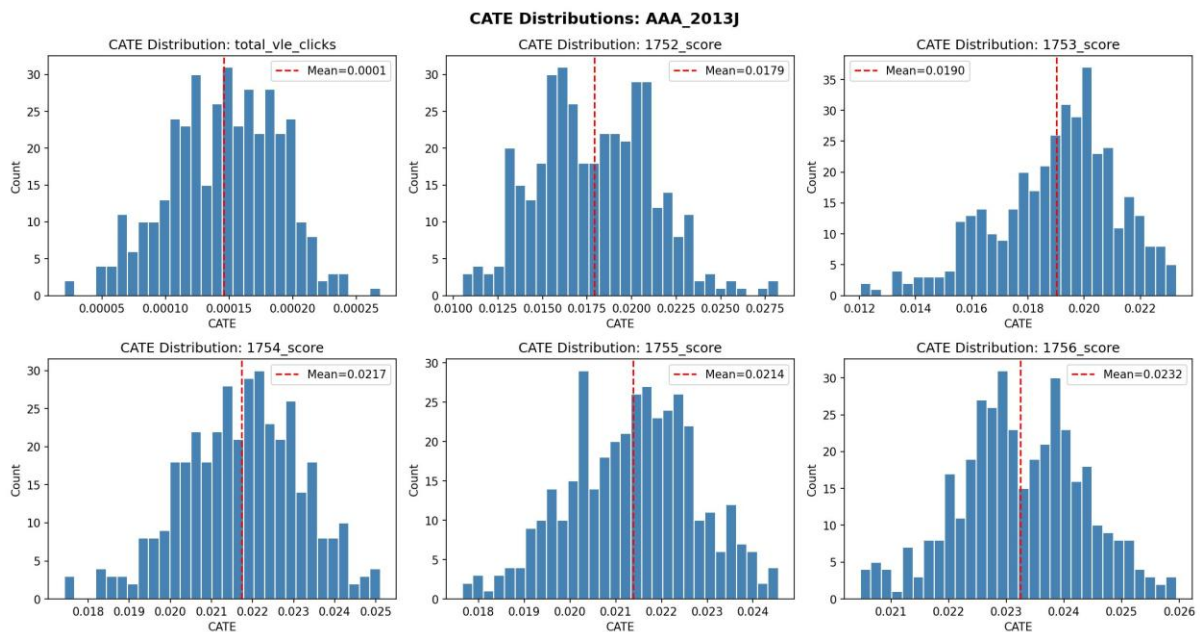


Figure 2: CATE distributions for all six treatments. Earlier assessments show wider, flatter distributions (high heterogeneity). Later assessments show tighter, more uniform distributions.

The distributions tell a clear story. For `total_vle_clicks`, the CATE is narrowly distributed around the mean — almost everyone is affected similarly. For early assessments (1752, 1753), the distribution is wide and flat, meaning some students are affected much more than others by their early performance. By 1754–1756, the distribution narrows considerably, meaning the treatment effect has become more uniform across students.

4.2 CATE Mean and Heterogeneity Trend

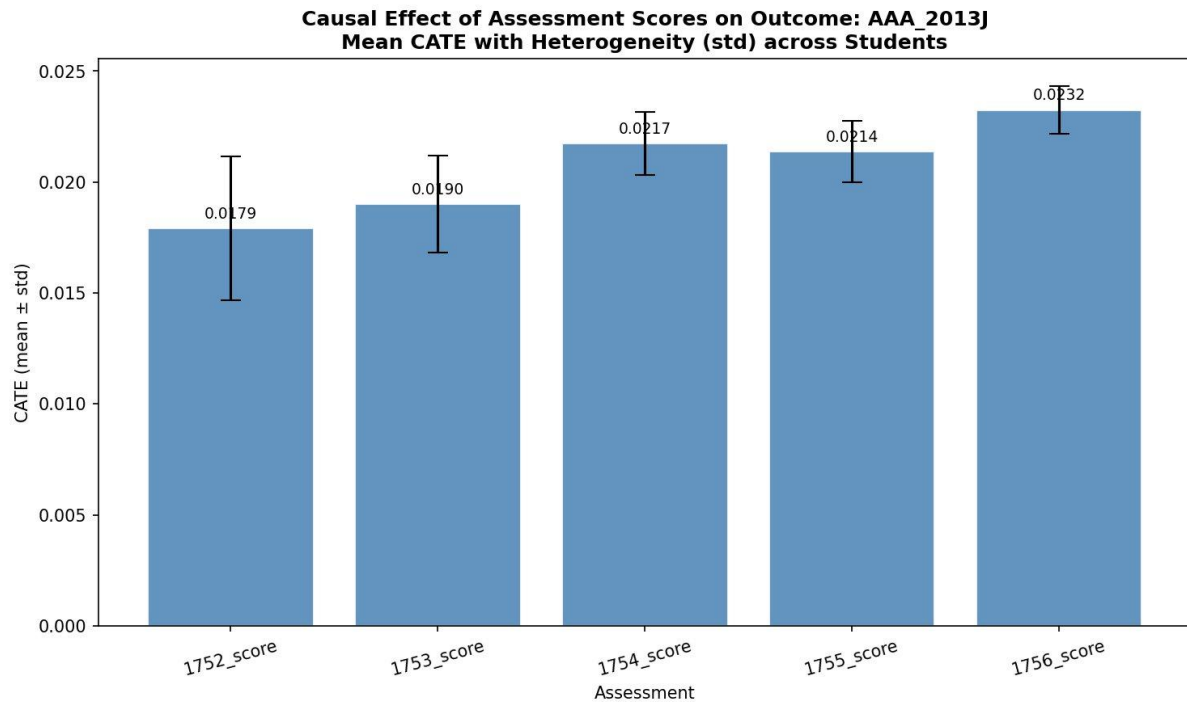


Figure 3: Mean CATE with heterogeneity (std) across assessments. Rising bars = stronger effect; shrinking error bars = less variation across students.

This chart summarises the key finding from the CATE stage. As the module progresses, both the strength and uniformity of the causal effect increase. By the final assessment (1756), the effect is the strongest and most consistent across all students — meaning it matters for everyone, regardless of background.

4.3 CATE vs Studied Credits

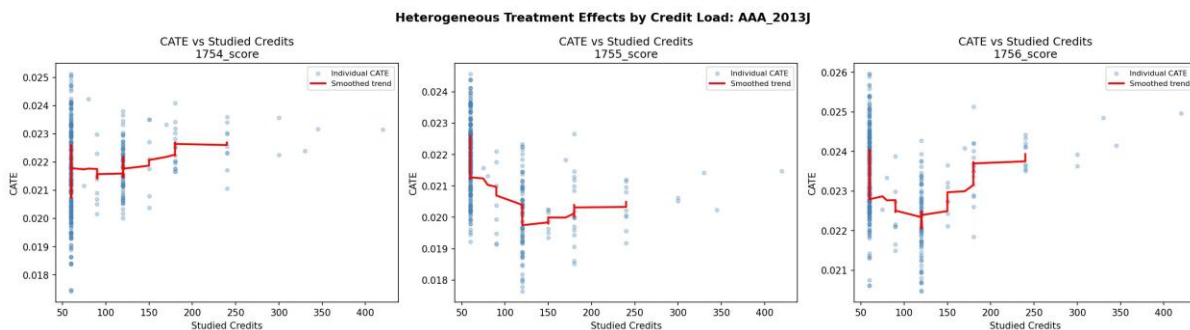


Figure 4: Individual CATE values vs studied credits for the three significant assessments (1754–1756). Red line = smoothed trend.

Most students are at 60 credits, with a smaller cluster at higher loads. For assessment 1755, there is a visible dip in CATE around 120 credits before recovering — students with moderate credit loads show slightly weaker treatment effects. For 1754 and 1756 the relationship is flatter. The sparse data at higher credit loads means these trends should be interpreted cautiously.

5. SHAP on CATE — Who Benefits Most?

KernelExplainer (SHAP) was applied to each fitted CausalForestDML model to identify which student characteristics drive treatment effect heterogeneity. SHAP values here represent each feature's contribution to an individual student's CATE relative to the mean — not to the outcome directly.

5.1 VLE Clicks

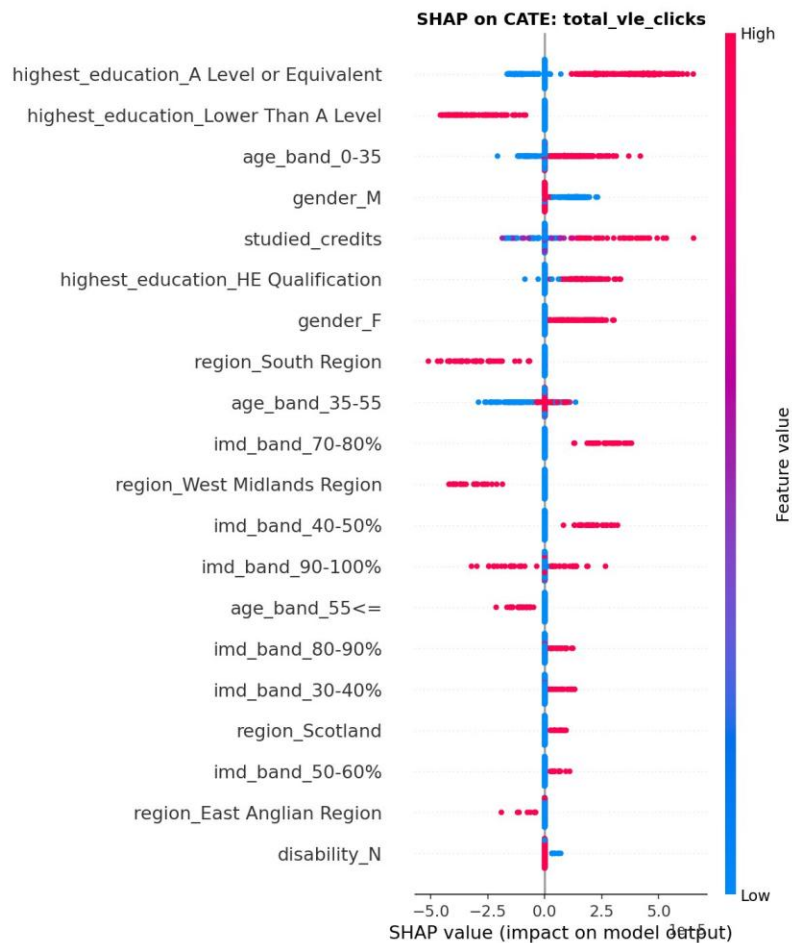


Figure 5: SHAP on CATE for total_vle_clicks. Top drivers: education level, age, and credit load.

For VLE engagement, education level is the dominant heterogeneity driver. Students with A-Level or equivalent qualifications (red = high feature value) tend to benefit more from online engagement. Students with lower than A-Level education show negative SHAP values — they benefit less from clicks. Younger students (0-35) and those with higher credit loads also show slightly higher CATE from VLE activity.

5.2 Assessment 1752 (Earliest)

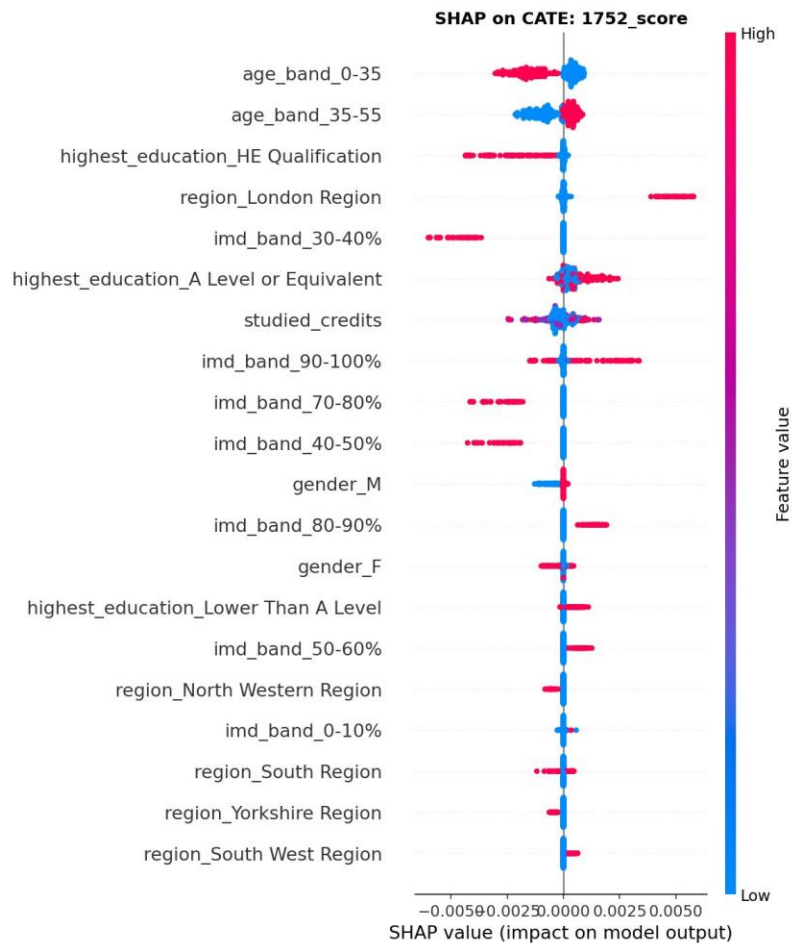


Figure 6: SHAP on CATE for assessment 1752. Age band is the dominant driver.

At the earliest assessment, age band is the primary heterogeneity driver. The age_band_35-55 group (mid-age students, shown in red/high) shows positive SHAP values — their early assessment performance has a stronger causal link to final outcome. Younger students (0-35, shown in blue/low) show weaker early assessment effects. This might reflect that mature students are more likely to stay on a consistent trajectory from early on, while younger students have more room to recover from early poor performance.

5.3 Assessment 1753

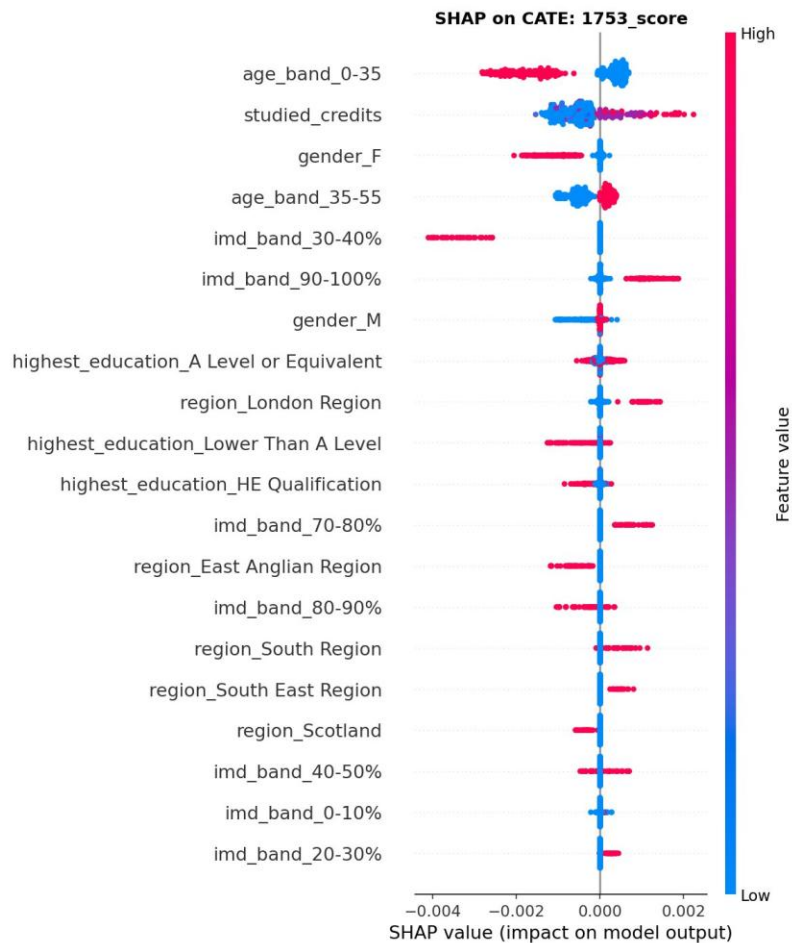


Figure 7: SHAP on CATE for assessment 1753. Age and credit load co-dominate.

By the second assessment, `age_band_0-35` moves to the top but now shows negative SHAP values — younger students' performance on this assessment is less predictive of their final outcome. Studied credits and `gender_F` emerge as secondary drivers. Female students show slightly positive SHAP values here, suggesting their 1753 performance is somewhat more predictive of outcome than male students at this stage.

5.4 Assessment 1754 (Mid-module)

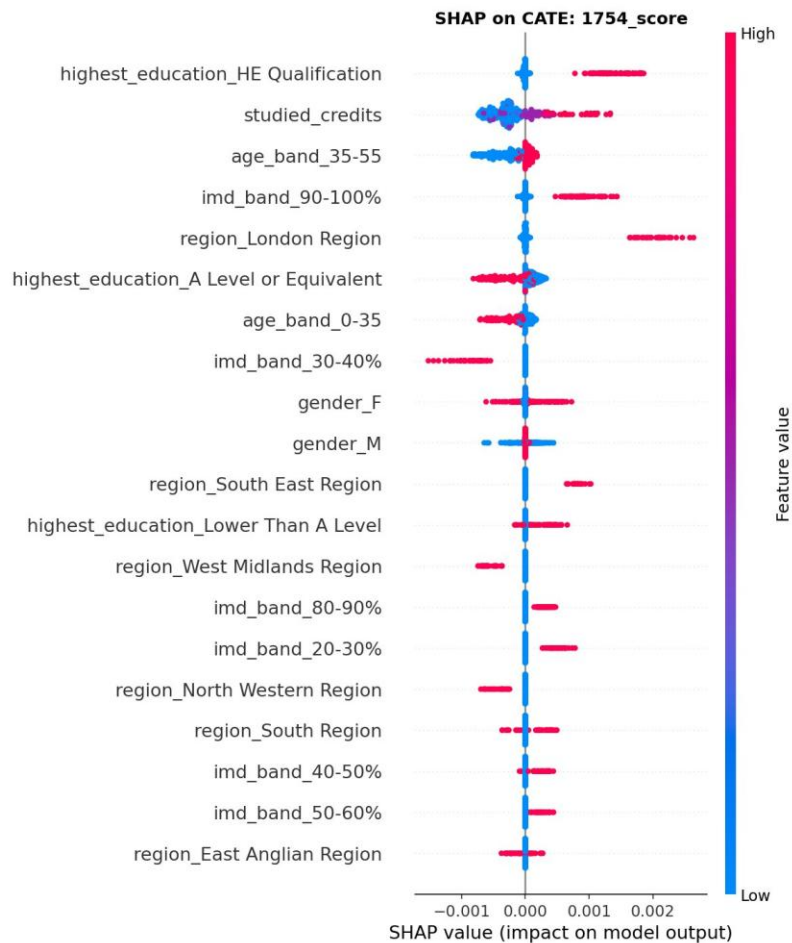


Figure 8: SHAP on CATE for assessment 1754. Education level takes over as top driver.

A clear shift happens at 1754 — `highest_education_HE Qualification` becomes the top driver, displacing age. Students with HE-level qualifications show higher CATE from mid-module performance, possibly because they have the academic foundations to capitalise on good mid-module results and convert them into Pass or Distinction outcomes. The London Region also appears as an unexpectedly strong positive driver here — worth flagging as an area for further investigation.

5.5 Assessment 1755

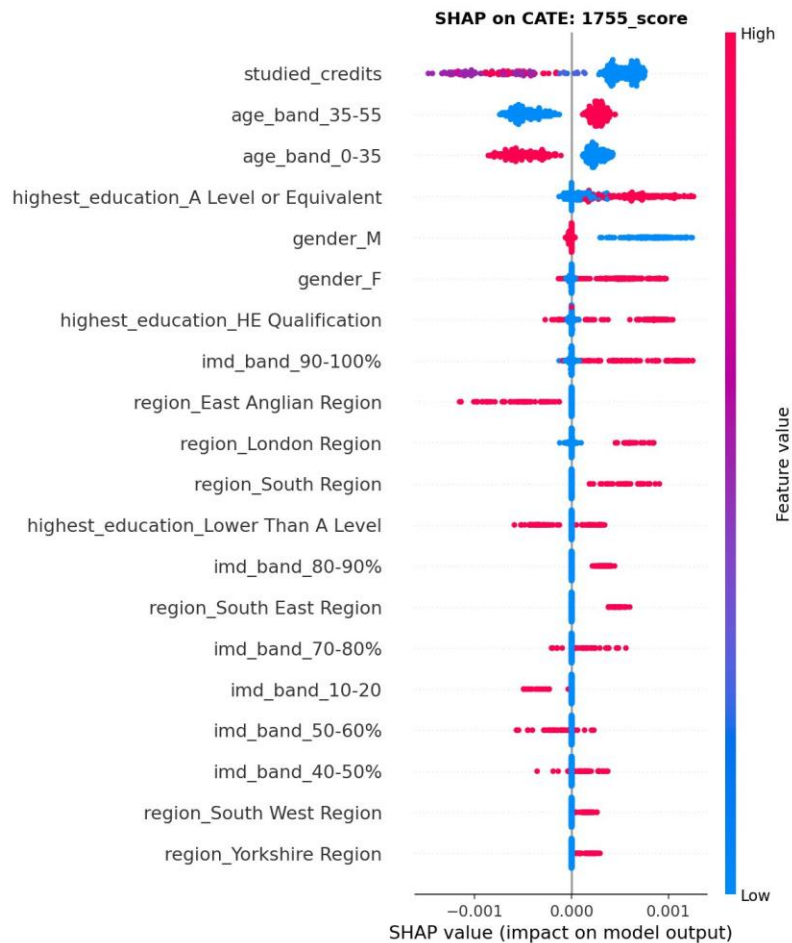


Figure 9: SHAP on CATE for assessment 1755. Credit load consolidates as top driver.

By assessment 1755, `studied_credits` becomes the top driver. Students with higher credit loads (red = high value) show positive SHAP values — their 1755 performance is more strongly tied to their final outcome. This makes intuitive sense: students carrying heavier workloads are under more pressure, so their performance on penultimate assessments is a stronger signal of whether they will Pass, Fail, or Withdraw.

5.6 Assessment 1756 (Final)

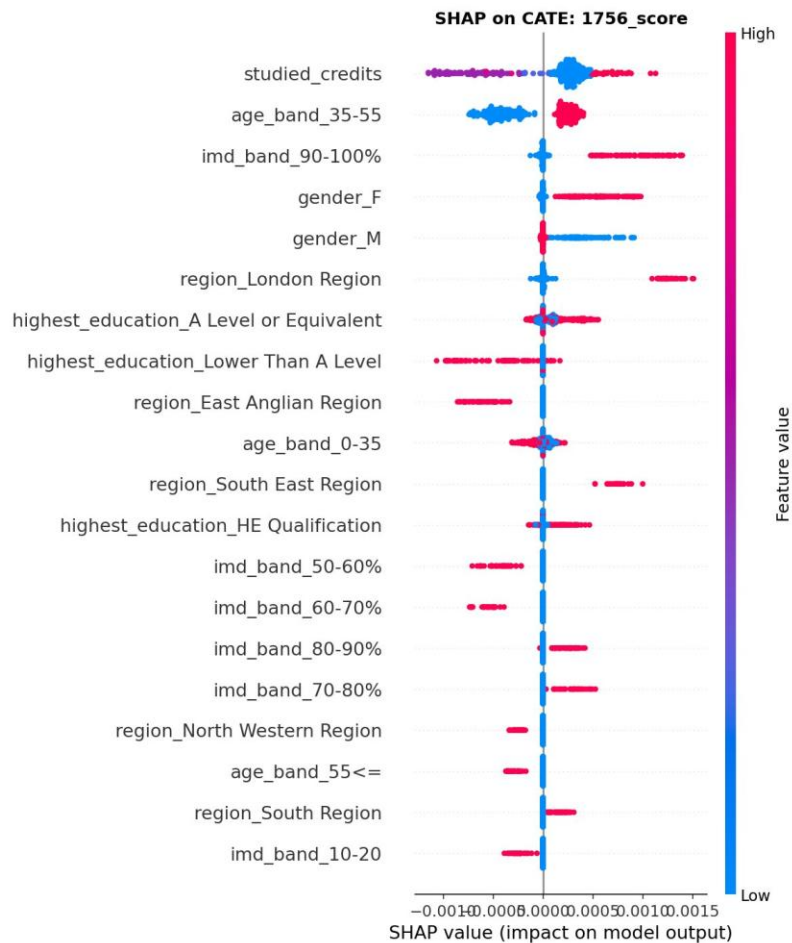


Figure 10: SHAP on CATE for assessment 1756. Credit load and age_band_35-55 dominate; gender divergence increases.

At the final assessment, studied_credits and age_band_35-55 consolidate as the two dominant drivers. Gender heterogeneity also increases noticeably — both gender_F and gender_M appear in the top features, suggesting male and female students diverge in how strongly their final assessment performance determines their outcome category. This is worth exploring further when scaling to other modules.

6. Summary of Findings

The central finding across all three stages is a progressive strengthening of causal effects across the module timeline:

1. VLE engagement has a small but real causal effect on outcome, uniform across all student types. It is not a strong differentiator between Pass and Distinction, but it does matter at the margins between Withdrawn and Fail.
2. Early assessment scores (1752, 1753) have weaker and more heterogeneous causal effects — students who perform poorly early are not necessarily locked into Withdrawn or Fail outcomes. Age is the main moderator at this stage.
3. Mid-to-late assessment scores (1754–1756) have stronger and more uniform effects — by this stage, performance is a near-universal determinant of whether a student achieves Pass/Distinction versus Fail/Withdrawn. Education level and credit load are the main moderators.
4. The heterogeneity transition across the module — from age-driven to credit-load-driven — is the most novel finding and the strongest candidate for the paper's central contribution when scaled across all 22 modules.

Limitations: AAA_2013J has only 383 students with 20 in the Distinction class, making findings for that category less stable. Results need to be replicated across the remaining 21 module-presentation combinations before drawing broader conclusions.